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ONE-HANDED RATCHET CLAMP TOOL

Abstract

A one-handed ratchet clamp tool includes a slide rod, a fixed arm, a movable arm, an operation lever, a clam member, and a release member. The fixed arm is disposed on one end of the slide rod. The movable arm has one end slidably mounted around the slide rod, and another end having an imaginary face and a ratchet which include a dip angle larger than 30 degrees. The operation lever is pivotally disposed on the movable arm. The clamp member is movably disposed on the operation lever, which optionally presses the clamp member, clamping or releasing the clamp member and the fixed arm. The release member is connected with the operation lever and has a pawl, which is engaged with or disengaged from the ratchet to position the operation lever or allow the operation lever to sway away from the movable arm. Convenience of one-handed operation is facilitated.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to ratchet clamp tools, and more particularly, to a one-handed ratchet clamp tool.

2. Description of the Related Art

[0002] Referring to CN209566035U, a Ratchet F Clamp is disclosed, which comprises a slide rod, a fixed clamp, and a moving clamp. The fixed clamp is fixed to one end of the slide rod, and the moving clamp is slidably mounted around the slide rod. The moving clamp comprises a moving clamp head and a clamping lever. The clamping lever is configured to be optionally pressed against the moving clamp, so that the moving clamp is clamped with respect to the fixed clamp. Therein, the moving clamp and the clamping lever comprises a ratchet and a pawl, respectively. The pawl is engaged with the ratchet through the elastic force of an elastic element, so as to position the clamping lever, thereby maintaining a clamping status. When a release handle on the pawl is operated, the pawl is disengaged with the ratchet, so as to release the clamped object.

[0003] In FIG. **3** of the patent above, the ratchet F clamp is illustrated in a releasing status, wherein the clamping lever is not pressed against the moving clamp. Because the moving clamp has a certain length, and the distribution angle included by the ratchet on the moving clamp and a clamping face ranges between 20 to 70 degrees; in other words, the ratchet is disposed to include a 20-degree dip angle with the clamping face. Therefore, an operation angle of the clamping lever with respect to the slide rod is relatively large, so that the user is unable to operate the clamping lever by using only one hand. Further, when two positioning pins on the moving clamp are in the clamping status, the moving clamp remains being fixed on the slide rod. However, after a long period of usage, wearing might be caused to deteriorate the clamping force thereof.

SUMMARY OF THE INVENTION

[0004] To improve the issues above, the present invention discloses a one-handed ratchet clamp tool, which comprises a ratchet having a dip angle ranging from 30 to 60 degrees, so as to reduce the burden of hand in operation, such that the user is allowed to carry out the clamping operation by using only one hand to hold an operation lever.

[0005] For achieving the aforementioned objectives, an embodiment of the present invention provides a one-handed ratchet clamp tool, comprising a slide rod, a fixed arm, a movable arm, an operation lever, a clamp member, and a release member. The fixed arm is disposed on one end of the slide rod, and comprises a first clamping face. The movable arm comprises a first end and a second end. The first end is mounted around the slide rod to slide with respect thereto. The second end comprises an imaginary face and a ratchet disposed away from the first clamping face, wherein the ratchet and the imaginary face include a dip angle larger than 30 degrees. The operation lever is pivotally disposed on the second end of the movable arm. The clamp member is movably disposed on the operation lever, and comprises a second clamping face disposed facing the first clamping face. The operation lever is configured to optionally press the clamp member, whereby the first clamping face and the second clamping face clamp or release together. The release member is connected with the operation lever, and comprises a pawl. The release member is operated between an engagement position and a disengagement position. At the engagement position, the pawl is engaged with the ratchet to position the operation lever. At the disengagement position, the pawl is disengaged from the ratchet, allowing the operation lever to sway away from the movable arm.

[0006] With such configuration, the position of the operation lever on the movable arm is biased toward the slide rod, achieving a more ergonomic design, which makes natural movement of the hand smoother and easier. The user does not need to excessively force or adjust the hand posture

when using only one hand, so that the convenience of one-handed operation is facilitated.
[0007] In another embodiment of the present invention, the first end of the movable arm comprises a slide hole to be mounted around the slide rod, with a wear-resistant abutting device disposed between the movable arm and the slide rod. Therefore, the relative position of the movable arm and the slide rod would not change due to wearing, so as to maintain the stable clamping effect. Also, the wear-resistant abutting device is removably disposed between the movable arm and the slide rod, facilitating the convenience of replacement and maintenance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] FIG. **1** is a perspective view of the one-handed ratchet clamp tool in accordance with an embodiment of the present invention.
[0009] FIG. **2** is an exploded view of the one-handed ratchet clamp tool in accordance with an embodiment of the present invention.
[0010] FIG. **3** is a side view of the one-handed ratchet clamp tool in accordance with an embodiment of the present invention.
[0011] FIG. **4** is a partially enlarged view of FIG. **3**.
[0012] FIG. **5** is a sectional view of the one-handed ratchet clamp tool in accordance with an embodiment of the present invention.
[0013] FIG. **6** is a partially enlarged view of FIG. **5**.
[0014] FIG. **7** is a schematic view of the operation status of the present invention, illustrating the operation lever and the slide rod being held by one hand.
[0015] FIG. **8** is a schematic view of the present invention, illustrating the operation lever being pressed by one hand.
[0016] FIG. **9** is a sectional view of FIG. **8**.
[0017] FIG. **10** is an operational schematic view of the release member being pressed.
[0018] FIG. **11** is a perspective view of the one-handed ratchet clamp tool in accordance with another embodiment of the present invention.
DETAILED DESCRIPTION OF THE INVENTION
[0019] The aforementioned and further advantages and features of the present invention will be understood by reference to the description of the preferred embodiment in conjunction with the accompanying drawings where the components are illustrated based on a proportion for explanation but not subject to the actual component proportion.
[0020] Referring to FIG. **1** to FIG. **10**, the present invention provides a one-handed ratchet clamp tool **100**, comprising a slide rod **10**, a fixed arm **20**, a movable arm **30**, an operation lever **40**, a clamp member **50**, and a release member **60**. The fixed arm **20** and the movable arm **30** are disposed on the same side of the slide rod **10**, respectively, so as to form an F-shaped overall structure.
[0021] The slide rod **10** has a cross-sectional face shaped in a long rectangular shape. The slide rod **10** comprises two wide lateral sides **11** and a first narrow lateral side **12** and a second narrow lateral side **13** arranged between the two wide lateral sides **11**.
[0022] The fixed arm **20** is disposed on one end of the slide rod **10**, with a first clamping face **21** disposed on one end of the fixed arm **20**. In the embodiment, the fixed arm **20** is fittingly mounted around the slide rod **10** through a sleeve hole **22**.
[0023] The movable arm **30** comprises a first end **31** and a second end **32**. The first end **31** comprises a slide hole **33** to be mounted around the slide rod **10**, so as to slide with respect to the slide rod **10**. As shown by FIG. **6**, the movable arm **30** comprises a first lock hole **34** and a second lock hole **35** disposed on two sides of the slide hole **33**, respectively. The first lock hole **34** is

arranged in parallel to the slide hole **33**. The second lock hole **35** is in transverse communication with the slide hole **33**. Therein, the first lock hole **34** is in communication with a first housing space **341**, and the second lock hole **35** is in communication with a second housing space **351**.

[0024] Referring to FIG. **2** and FIG. **3**, the second end **32** defines an imaginary face **36** and comprises a ratchet **37** arranged on one side in opposite to the first clamping face **21**. The ratchet **37** comprises at least 10 teeth. The ratchet **37** and the imaginary face **36** include a dip angle A larger than 30 degrees. In the embodiment, the dip angle A is 50 degrees. When the present invention is used to clamp an object, the imaginary face **36** is approximately in parallel to the surface of the object. Also, the second end **32** comprises an open recess **38** facing the first clamping face **21**.

[0025] The operation lever **40** is pivotally disposed on the second end **32** of the movable arm **30**. In the embodiment, the operation lever **40** is pivotally disposed on the second end **32** of the movable arm **30** along a pivot shaft **70**. The movable arm **30** defines a longitudinal axle L which is perpendicular to the imaginary face **36** and passes through an axle center X of the pivot shaft **70**. Therein, the longitudinal axle L is in parallel to a length direction of the slide rod **10**. A distance between the second narrow lateral side **13** of the slide rod **10** and the operation lever **40** is larger than a distance between the first narrow lateral side **12** and the operation lever **40**.

[0026] The operation lever **40** comprises a pressing part **41** formed in a concave arc shape, and a pivot hole **42** and a protrusion part **43** away from the pressing part **41**. The pressing part **41** is biased from the longitudinal axle L toward the slide rod **10**. The aforementioned pivot shaft **70** passes through the pivot hole **42** and is held by the recess **38**, allowing the operation lever **40** to pivotally sway with respect to the movable arm **30**. Referring to FIG. **2** and FIG. **4**, the operation lever **40** comprises a non-circular assembly bore **44** in adjacent to the pivot hole **42**, and the assembly bore **44** comprises an abutting wall **441**.

[0027] The clamp member **50** is movably disposed on the operation lever **40**. The clamp member **50** comprises a second clamping face **51** disposed in positional alignment with the first clamping face **21**. In the embodiment, the clamp member **50** comprises a long groove **52** combined with the aforementioned pivot shaft **70**, such that the clamp member **50** and the operation lever **40** are connected on the movable arm **30**, and the clamp member **50** is allowed to pivotally sway with respect to the movable arm **30**. The operation lever **40** optionally presses the clamp member **50** through the protrusion part **43**, such that the first clamping face **21** and the second clamping face **51** clamp or release together.

[0028] Referring to FIG. **2**, FIG. **4**, and FIG. **10**, the release member **60** is connected with the operation lever **40**. The release member **60** comprises a pawl **61**. The release member **60** is allowed to be operated between an engagement position and a disengagement position. At the engagement position, the pawl **61** is engaged with the ratchet **37**, so as to position the operation lever **40**. At the disengagement position, the pawl **61** is disengaged from the ratchet **37**, so that the operation lever **40** is allowed to sway away from the movable arm **30**, thereby releasing the object.

[0029] Therein, the release member **60** has an elastic arm **62** extending therefrom to abut against the operation lever **40**, so as to be maintained at the engagement position, such that the operation lever **40** is in a positioned status. As shown by FIG. **4**, the release member **60** comprises a non-circular convex shaft **63**, which is disposed in the assembly bore **44** and biased by a small angle. The convex shaft **63** comprises a first contact face **631** and a second contact face **632**. The first contact face **631** and the second contact face **632** are not arranged on a same plane, wherein the first contact face **631** and the second contact face **632** are biased by an angle which is 15 degrees or less to facilitate the pressing operation. Therefore, at the engagement position as shown by FIG. **4**, the first contact face **631** abuts against the abutting wall **441** of the assembly bore **44**. And, as shown by FIG. **10**, when the release member **60** is pressed by a force to be switched from the engagement position to the disengagement position, the pawl **61** is disengaged from the ratchet **37**, enabling the operation lever **40** to pivotally sway away from the movable arm **30** for releasing the object. Therein, when the release member **60** is being pressed, the second contact face **632** abuts against the abutting wall **441**, so as to limit a bias angle of the release member **60** with respect to the operation lever **40**.

[0030] Referring to FIG. **3** and FIG. **7** to FIG. **9**, the dip angle A included by the ratchet **37** and the imaginary face **36** is larger than 30 degrees, such that the operation lever **40** is allowed to be biased toward the slide rod **10** on the movable arm **30**. With such configuration, the user is able to conveniently grip the operation lever **40** and the slide rod **10** by using one hand, so as to pull the operation lever **40**, whereby the clamp member **50** and the fixed arm **20** clamp the object together. Also, the minimum distance from the pressing part **41** to the slide rod **10** ranges from 80 mm to 100 mm, which is adapted to the largest width of the expanding palm of the user, facilitating the one-hand operation by the user.

[0031] In another embodiment of the present embodiment, a wear-resistant abutting device **80** is disposed between the movable arm **30** and the slide rod **10**, so as to prevent the slide wearing between the movable arm **30** and the slide rod **10** after a long-time usage from affecting the clamping effect. Therein, referring to FIG. **6**, the wear-resistant abutting device **80** comprises a first wear-resistant member **81** and a second wear-resistant member **82**. The first wear-resistant member **81** is removably disposed between the movable arm **30** and the first narrow lateral side **12**. The second wear-resistant member **82** is removably disposed between the movable arm **30** and the second narrow lateral side **13**.

[0032] The first wear-resistant member **81** is fastened to the first lock hole **34** through a first fastening member **83**, and the second wear-resistant member **82** is fastened to the second lock hole **35** through a second fastening member **84**. Also, the first wear-resistant member **81** is positioned in the first housing space **341** and not exposed from the bottom edge of the movable arm **30**. The first wear-resistant member **81** is formed in a rectangular shape and has one lateral side thereof abutting against the first narrow lateral side **12**. The second wear-resistant member **82** is positioned in the second housing space **351** and not exposed from the top edge of the movable arm **30**. The second wear-resistant member **82** is formed in a rectangular shape and has one lateral side thereof abutting against the second narrow lateral side **13**. Therefore, when each abutting lateral side of the first wear-resistant member **81** and the second wear-resistant member **82** is worn out, each of the first wear-resistant member **81** and the second wear-resistant member **82** is allowed to be rotated by detaching the first fastening member **83** and the second fastening member **84** to have another lateral side not yet worn out abut against the first narrow lateral side **12** and the second narrow lateral side **13**, respectively, to facilitate continuous usage. When four lateral sides of the first wear-resistant member **81** and the second wear-resistant member **82** are all worn out, the original first wear-resistant member **81** and second wear-resistant member **82** can be replaced with new ones.

[0033] In another embodiment of the present invention, a positioning device **90** is disposed between the second narrow lateral side **13** of the slide rod **10** and the movable arm **30**. With the positioning device **90** abutting against the second narrow lateral side **13**, the movable arm **30** is allowed to be effectively engaged and positioned on the slide rod **10**.

[0034] Referring to FIG. **6**, the positioning device **90** comprises a combining member **91** and a bead member **92**. The combining member **91** is a set screw, which is screwed to a thread bore **30***b* of the movable arm **30**. The bead member **92** is rotatably disposed on an end part of the combining member **91** and abuts against the second narrow lateral side **13**, whereby the movable arm **30** is engaged and positioned on the slide rod **10**.

[0035] In another embodiment of the present invention, a grip **30***a* is further included. The grip **30***a* is removably mounted around the first end **31** of the movable arm **30**, providing an optimal gripping comfort and stability for the user during clamping operation. Further, to prevent the detachment of the movable arm **30** from the slide rod **10**, a fixing screw member **14** is fixed to one end of the slide rod **10** away from the fixed arm **20**, thereby preventing the movable arm **30** from leaving the slide rod **10** toward the direction away from the fixed arm **20**.

[0036] Referring to FIG. **11**, in another embodiment of the present invention, the slide rod **10** and the fixed arm **20** are integrally formed. In other words, the fixed arm **20** and the slide rod **10** form an L-shaped rod member.
[0037] With the foregoing configuration, with an optimal design of the dip angle A of the ratchet **37**, the operation lever **40** is biased toward the slide rod **10** on the movable arm **30**, facilitating a one-handed and ergonomic operation, so as to lower the burden on the hand of the user.
[0038] Also, with the wear-resistant abutting device **80** removably disposed between the movable arm **30** and the slide rod **10**, the relative position of the movable arm **30** with respect to the slide rod **10** is prevented from changing due to wearing, so as to maintain the clamping effect of the clamp tool and facilitate the convenience of maintenance and parts replacement.
[0039] Although particular embodiments of the invention have been described in detail for purposes of illustration, various modifications and enhancements may be made without departing from the spirit and scope of the invention. Accordingly, the invention is not to be limited except as by the appended claims.

Claims

1. A one-handed ratchet clamp tool, comprising: a slide rod; a fixed arm disposed on one end of the slide rod and comprising a first clamping face; a movable arm comprising a first end and a second end, the first end mounted around the slide rod and slidable with respect to the slide rod, the second end defining an imaginary face and comprising a ratchet arranged away from the first clamping face, the ratchet and the imaginary face including a dip angle larger than 30 degrees; an operation lever pivotally disposed on the second end of the movable arm; a clamp member movably disposed on the operation lever, the clamp member comprising a second clamping face arranged in positional alignment with the first clamping face, the operation lever optionally pressing the clamp member, whereby the first clamping face and the second clamping face clamp or release together; and a release member connected with the operation lever and comprising a pawl, the release member being operated between an engagement position and a disengagement position; at the engagement position, the pawl being engaged with the ratchet to position the operation lever; at the disengagement position, the pawl being disengaged from the ratchet, allowing the operation lever to sway away from the movable arm.
2. The one-handed ratchet clamp tool of claim 1, wherein the operation lever is pivotally disposed on the second end along a pivot shaft; the movable arm defines a longitudinal axle which is perpendicular to the imaginary face and passes through an axle center of the pivot shaft.
3. The one-handed ratchet clamp tool of claim 2, wherein the dip angle is 50 degrees; the longitudinal axle is in parallel to a length direction of the slide rod.
4. The one-handed ratchet clamp tool of claim 2, wherein the operation lever comprises a pressing part biased from the longitudinal axle toward the slide rod, and a pivot hole and a protrusion part arranged away from the pressing part; the second end comprises an open recess facing the first clamping face; the pivot shaft passes through the pivot hole and is held by the recess; the protrusion part presses the clamp member.
5. The one-handed ratchet clamp tool of claim 4, wherein the clamp member comprises a long groove combined with the pivot shaft.
6. The one-handed ratchet clamp tool of claim 4, wherein a minimum distance from the pressing part to the slide rod ranges from 80 mm to 100 mm.
7. The one-handed ratchet clamp tool of claim 2, wherein the release member comprises an elastic arm extending to abut against the operation lever, so as to be maintained at the engagement position.
8. The one-handed ratchet clamp tool of claim 7, wherein the operation lever comprises a non-circular assembly bore; the release member comprises a non-circular convex shaft, which is

disposed in the assembly bore.

9. The one-handed ratchet clamp tool of claim 8, wherein the assembly bore comprises an abutting wall; the convex shaft comprises a first contact face and a second contact face; the first contact face and the second contact face are not arranged on a same plane; at the engagement position, the first contact face abuts against the abutting wall; at the disengagement position, the second contact face abuts against the abutting wall.

10. The one-handed ratchet clamp tool of claim 1, wherein the first end of the movable arm comprises a slide hole to be mounted around the slide rod; a wear-resistant abutting device is disposed between the movable arm and the slide rod.

11. The one-handed ratchet clamp tool of claim 10, wherein the slide rod comprises two wide lateral sides, and a first narrow lateral side and a second narrow lateral side arranged between the two wide lateral sides; a distance between the second narrow lateral side and the operation lever is larger than a distance between the first narrow lateral side and the operation lever; the wear-resistant abutting device comprises a first wear-resistant member and a second wear-resistant member; the first wear-resistant member is removably disposed between the movable arm and the first narrow lateral side; the second wear-resistant member is removably disposed between the movable arm and the second narrow lateral side.

12. The one-handed ratchet clamp tool of claim 11, wherein the movable arm comprises a first lock hole and a second lock hole disposed on two sides of the slide hole, respectively; the first lock hole is arranged in parallel to the slide hole, the second lock hole is in transverse communication with the slide hole; the first wear-resistant member is fastened to the first lock hole through a first fastening member, and the second wear-resistant member is fastened to the second lock hole through a second fastening member.

13. The one-handed ratchet clamp tool of claim 11, wherein a positioning device is disposed between the second narrow lateral side and the movable arm.

14. The one-handed ratchet clamp tool of claim 13, wherein the positioning device comprises a combining member and a bead member; the combining member is screwed to the movable arm; the bead member is rotatably disposed on an end part of the combining member and abuts against the second narrow lateral side.

15. The one-handed ratchet clamp tool of claim 10, further comprising a grip removably mounted around the first end of the movable arm.